

EXPERIMENT F-20 THE PONTENTIOMETER

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NAME: SHISAMA SHIVKA AITON

REG NO: F16/1978/2022

PARTNERS: JOSHUA OMBUI - F16/1983/2022
YVONNE ATIENO - F16/1976/2022

OBJECTIVES

To determine the E_{mf} of a dry cell
Explain and apply the laws of thermodynamics to solve problems related to energy conversions in thermodynamic systems.

APPARATUS

1. A potentiometer with sliding contact
2. Accumulator (driving voltage)
3. Galvanometer
4. Switch
5. A variable resistor
6. An ammeter

THEORY

The potentiometer is a device for measuring/ comparing potential differences.

~~Its~~ Its essential features are:

- (i) A steady source of potential known as 'driver cell', D .
- (ii) A resistor AB of uniform resistance of $r \Omega \text{ cm}^{-1}$ through which the accumulator sends a steady current (I).

The potential difference (p.d.) between point A and any other point (J) along AB of the wire is proportional to the length AJ and is given by:

$$V_{AJ} = I \times (\text{resistance of AJ}) = I \times \frac{\rho L}{A} \dots (20.1)$$

Where ρ = resistivity of the wire

A = cross sectional area of the wire

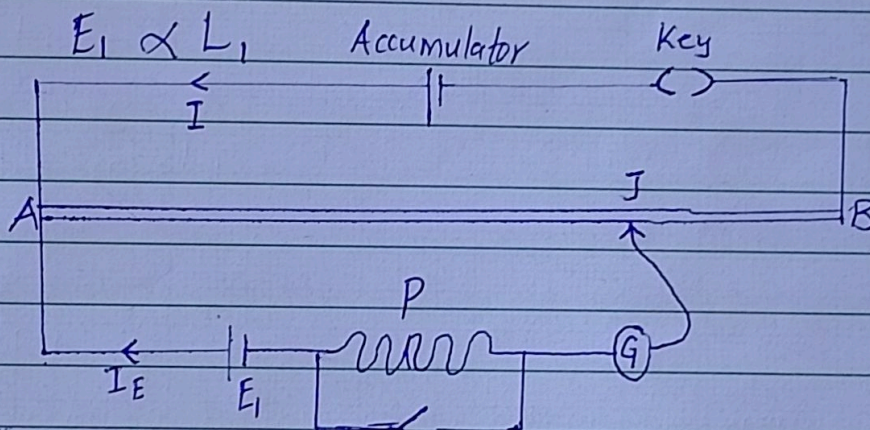
L = length AJ

Hence p.d. (V) $\propto L$ since I , ρ and A are constant along the wire.

If a cell of e.m.f., E_1 (E_1 is p.d. between A and B) is now connected as shown below, a position for J can be found for which the sensitive galvanometer (G) shows no deflection (the potentiometer is balanced). Thus:

$$E_1 = \text{p.d. across AJ} = I \cdot L_1 (r)$$

OR



Cell E_1 drives a current (I_E) in the opposite direction to that of current I .

If ~~now~~ E_1 is replaced by a cell of known e.m.f. (E_2) then and a new balance point, L_2 can be found such that;

$$E_2 \propto L_2$$

Combining these equations: $\frac{E_1}{E_2} = \frac{L_1}{L_2}$ or $E_1 = \frac{L_1}{L_2} E_2$

(20.2)

This is the principle of a potentiometer and the e.m.f. of an ~~unkor~~ unknown cell (E_1) can be found using Equation (20.2) if E_2 is known.

METHOD

1. The circuit was connected as shown above with the standard cell in place of E_1 . Both positive poles of the accumulator and standard cell were connected to A and the circuit was checked by the lecturer in charge.
2. The driver cell circuit was switched on and J adjusted until a position of no deflection was found in the galvanometer.
~~So~~ The protective resistor (P) was switched out and final adjustments were made. The balance length AJ was measured.
3. The connections ~~to~~ of the driver cell ~~and~~ were reversed and again the balance length AJ measured. The mean balance length was calculated. Readings were recorded in the datasheet.
4. The procedure was repeated for the Leclanche cell and the Daniell cell, noting the lengths required to obtain the balance.
The cells were assembled and correct liquids added to them.
5. All the measurements were repeated at least once to obtain mean values for the lengths.

EXPERIMENT F-20

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THE POTENTIOMETER

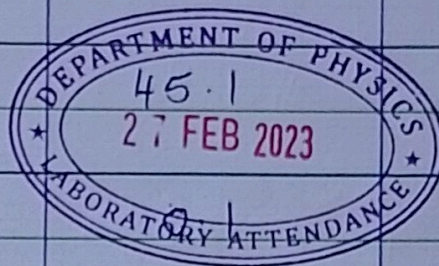
NAME: A SHISAMA SHIVKA AITON

REG. NO: F16/1978/2022

PARTNERS: OMBUI JOSHUA - F16/1983/2022
YVONNE ATIENO - F16/1976/2022

DATA SHEET

	Standard cell	Leclanche cell	Daniel cell
Balance length (cm)	35.8	47.7	51.4
On reversing (cm)	47.6	42.5	42.4
Mean length (cm)	41.7	45.1	46.9
Setting error $\pm$ (cm)	0.1	0.1	0.1
e.m.f. of cell (V)	1.012	0.9357	0.8998
Error in e.m.f $\pm$ (V)	0.001	0.005243	0.004965



CALCULATIONS

The e.m.f's of the Leclanche and Daniell cells were calculated as below:

Leclanche cell

$$E_1 = \frac{L_1}{L_2} E_2$$

$$= \frac{41.7}{45.1} \times 1.012 = 0.9357 \text{ V}$$

Error in e.m.f:

$$= \left[\left(\frac{0.1}{41.7} + \frac{0.1}{45.1} \right) 0.9246 \right] = \frac{868}{188067} \times 0.9246 = \pm 4.267 \times 10^{-3}$$

$$\pm \left(\frac{4.267 \times 10^{-3}}{0.9246} + \frac{0.001}{1.012} \right) 0.9357 = 5.604 \times 10^{-3} \times 0.9357$$
$$= \pm 5.243 \times 10^{-3} \text{ V}$$

$$\therefore \underline{E_1 = 0.9357 \pm 0.005243 \text{ V}}$$

Daniell cell

$$E_1 = \frac{L_1}{L_2} E_2$$

$$= \frac{41.7}{46.9} \times 1.012 = 0.8998 \text{ V}$$

Error in e.m.f:

$$= \left[\left(\frac{0.1}{41.7} + \frac{0.1}{46.9} \right) 0.8891 \right] + \frac{0.001}{1.012} \times 0.8998$$

$$= \left(\frac{886}{195573} + \frac{1}{1012} \right) 0.8998 = \pm 4.965 \times 10^{-3} \text{ V}$$

$$\therefore \underline{E_1 = 0.8998 \pm 0.004965 \text{ V}}$$

ANALYSIS

The emf of the driver cell should be greater than that of the cell under test because the fall of potential across the potentiometer wire should not be less than the emf of the cell to be determined.

A potentiometer is considered as a voltmeter with an infinite resistance because the emf ~~area~~ of a cell can be measured using it and it does not draw any current from the cell.

The factors affecting the accuracy of the potentiometer are:

1. Internal temperature.
2. Consequent resistance change in the current carrying resistor.
3. Potential gradient.
4. Area of cross-section of the wire.
5. Current passing through potentiometer wire.

The objective of reversing the terminals of the cell was to prevent the resistance due to joule heat from being added to the measured resistance, ultimately preventing unwanted resistance.

The ~~accuracy~~ experimental results will not be accurate if the resistor is not switched out because it can change the current flow in other components in the circuit such as the capacitor.

DISCUSSIONS

From the results, the emf of Leclanche cell was found to be $0.9357 \pm 0.005243 \text{ V}$ which is close to the standard value of 1.4 volts.

Whereas, the emf of Daniell cell was found as $0.8998 \pm 0.004965 \text{ V}$ which compares to the standard value of about 1.0 volts.

The differences in the emfs are due to the factors affecting the accuracy of the potentiometer as listed on the analysis part.

It was also found ~~out~~ that the emf of the driver cell was greater than those of the Leclanche and Daniell cells.

CONCLUSION

The emf of the dry cell (Leclanche cell) was found to be $0.9357 \pm 0.005243 \text{ V}$ which is close to the standard value of 1.4 V.

The confidence level is 33% thus showing the objectives were not fully met.

REFERENCES

Internet: ~~quora.com~~

www.quora.com

<https://www.toppr.com>